

Medical AI Assistant — API Contracts

Overview

This document defines the REST API contracts for the AI-powered medical consultation assistant. The pipeline is: **Audio** → **Whisper STT** → **pyannote Diarization** → **LangChain Orchestration** → **Gemini Extraction** → **RAG (ChromaDB)** → **Rules Engine** → **Structured Note** → **Approval** → **Patient Summary**.

Base URL: `https://api.medassist.local/v1`

All requests require: `Authorization: Bearer <token>` and `Content-Type: application/json` unless noted.

1. Session Management

POST /sessions

Create a new consultation session.

Request

```
{
  "doctor_id": "d_001",
  "patient_id": "p_092",
  "language_hint": "kn-en"
}
```

Response **201**

```
{
  "session_id": "sess_7x9f2k",
  "status": "ready",
  "created_at": "2026-10-08T09:00:00Z",
  "patient_snapshot": {
    "name": "Ravi Kumar",
    "age": 52,
    "chronic_conditions": ["hypertension", "type2_diabetes"],
    "known_allergies": ["penicillin"]
  }
}
```

GET /sessions/{session_id}

Fetch live session state.

Response 200

```
{
  "session_id": "sess_7x9f2k",
  "status": "active",
  "duration_seconds": 340,
  "transcript_lines": 28,
  "warnings_fired": 1
}
```

PATCH /sessions/{session_id}/end

Close the session and trigger final note generation.

Response 200

```
{
  "session_id": "sess_7x9f2k",
  "status": "pending_approval",
  "note_id": "note_3m1pq8"
}
```

2. Audio Ingestion & Transcription

POST /sessions/{session_id}/audio

Upload an audio chunk for real-time transcription. Accepts streaming chunks (multipart) or a full file.

Content-Type: multipart/form-data

Form fields

Field	Type	Notes
audio	file	.wav or .webm, max 30s per chunk
chunk_index	int	Ordering index
is_final	bool	true on last chunk

Response 200

```
{
  "chunk_index": 4,
  "transcript": [
    {
      "speaker": "DOCTOR",
      "text": "BP kaisa hai?",
      "start_ms": 0,
      "end_ms": 1200,
      "confidence": 0.94
    },
    {
      "speaker": "PATIENT",
      "text": "Bahut zyada hai sir, 3 din se",
      "start_ms": 1400,
      "end_ms": 3600,
      "confidence": 0.89
    }
  ],
  "language_detected": "kn-en"
}
```

GET /sessions/{session_id}/transcript

Fetch the full diarized transcript for a session so far.

Response 200

```
{
  "session_id": "sess_7x9f2k",
  "lines": [
    { "speaker": "DOCTOR", "text": "...", "start_ms": 0 },
    { "speaker": "PATIENT", "text": "...", "start_ms": 1400 }
  ],
  "total_duration_ms": 18400
}
```

3. Real-Time Extraction (Gemini Flash)

POST /sessions/{session_id}/extract

Trigger mid-session extraction — called automatically by LangChain whenever a drug name or diagnosis keyword is detected.

Request

```
{
  "trigger": "drug_mention",
  "trigger_value": "amoxicillin",
  "transcript_window": "last_60s"
}
```

Response 200

```
{
  "extracted": {
    "diagnoses": ["acute_tonsillitis"],
    "medications_mentioned": ["amoxicillin"],
    "symptoms": ["sore throat", "fever for 3 days"],
    "vitals_mentioned": { "bp": "150/90" }
  },
  "rag_triggered": true,
  "rag_result": {
    "relevant_history": [
```

```

    {
      "date": "2025-06-14",
      "event": "Rash after amoxicillin – suspected allergy",
      "source": "consultation_note"
    }
  ],
  "warnings": [
    {
      "level": "critical",
      "code": "ALLERGY_CONFLICT",
      "message": "Patient had adverse reaction to amoxicillin on 2025-06-14. Prescribing this drug is contraindicated."
    }
  ]
}

```

4. RAG — Patient History

POST /patients/{patient_id}/history/embed

Embed and store a new history record (called after each approved consultation).

Request

```

{
  "type": "consultation_note",
  "date": "2026-10-08",
  "content": {
    "diagnosis": "hypertension",
    "medications": ["amlodipine 5mg"],
    "outcome": "stable",
    "notes": "Patient tolerates amlodipine well."
  }
}

```

Response 201

```
{
  "record_id": "rec_99xz",
  "embedded": true,
  "vector_id": "chroma_v_1023"
}
```

POST /patients/{patient_id}/history/query

RAG retrieval — fetch the most relevant history records for a given clinical context.

Request

```
{
  "query": "amoxicillin allergy drug reaction",
  "top_k": 5,
  "filters": {
    "types": ["consultation_note", "allergy_record", "lab_result"]
  }
}
```

Response 200

```
{
  "results": [
    {
      "record_id": "rec_44ab",
      "type": "allergy_record",
      "date": "2025-06-14",
      "relevance_score": 0.97,
      "summary": "Rash after amoxicillin – suspected allergy"
    }
  ]
}
```

5. Rules Engine

POST /rules/check

Run the rules engine against a set of extracted clinical data. Can be called independently of the LLM pipeline.

Request

```
{
  "patient_id": "p_092",
  "medications": ["amoxicillin", "metformin"],
  "diagnoses": ["acute_tonsillitis"],
  "vitals": { "bp": "150/90" },
  "session_id": "sess_7x9f2k"
}
```

Response 200

```
{
  "warnings": [
    {
      "level": "critical",
      "code": "ALLERGY_CONFLICT",
      "drug": "amoxicillin",
      "message": "Allergy record on file. Do not prescribe."
    },
    {
      "level": "advisory",
      "code": "DOSAGE_WATCH",
      "drug": "metformin",
      "message": "Patient has CKD markers. Confirm renal function before dosing."
    }
  ],
  "passed": false
}
```

Warning levels

Level	Meaning
critical	Block — must be resolved before note is approved
advisory	Warn — doctor must acknowledge to proceed
info	Note — surfaced in record, no action required

6. Note Generation & Approval

POST /notes/generate

Generate the structured clinical note (Gemini Pro). Called automatically on session end.

Request

```
{
  "session_id": "sess_7x9f2k",
  "mode": "full"
}
```

Response 200

```
{
  "note_id": "note_3m1pq8",
  "status": "draft",
  "content": {
    "chief_complaint": "High BP and sore throat for 3 days",
    "history_of_present_illness": "52-year-old male with known hypertension presenting with...",
    "diagnosis": [
      { "code": "I10", "label": "Essential hypertension" },
      { "code": "J03.90", "label": "Acute tonsillitis, unspecified" }
    ],
    "prescription": [
      {
        "drug": "Azithromycin",
        "dose": "500mg",
        "frequency": "once daily",
        "duration": "5 days",
        "route": "oral"
      }
    ],
    "follow_up": "Review in 7 days or earlier if symptoms worsen.",
    "doctor_notes": ""
  },
  "warnings_unresolved": 0
}
```

PATCH /notes/{note_id}

Doctor edits the draft note before approval.

Request

```
{
  "prescription": [
    {
      "drug": "Azithromycin",
      "dose": "500mg",
      "frequency": "once daily",
      "duration": "5 days",
      "route": "oral"
    }
  ],
  "doctor_notes": "Patient counselled on salt restriction."
}
```

Response 200 — returns updated note object.

POST /notes/{note_id}/approve

Doctor approves the note — triggers save to DB and email notification dispatch via Resend.

Request

```
{
  "doctor_id": "d_001",
  "signature_token": "eyJhb..."
}
```

Response 200

```
{
  "note_id": "note_3m1pq8",
  "status": "approved",
  "approved_at": "2026-10-08T09:28:00Z",
  "email_dispatch": {
    "queued": true,
    "recipient": "patient@example.com"
  }
}
```

7. Patient Notification (Resend Email)

POST /summaries/dispatch

Generate a simplified patient-facing summary and send via email using Resend.

Request

```
{
  "note_id": "note_3m1pq8",
  "language": "kn",
  "channel": "email",
  "recipient_email": "patient@example.com"
}
```

Response 200

```
{
  "dispatch_id": "disp_6t3w",
  "status": "sent",
  "resend_message_id": "msg_ABC123xyz",
  "subject": "Your consultation summary – Dr. Sharma, 8 Oct 2026",
  "message_preview": "ನಿಮ್ಮ ಇಂದಿನ ಸಂಪರ್ಕದ ಸಾರಾಂಶ: ರೋಗನಿರ್ಣಯ – ಗಂಟಲು ಸೋಂಕು. ಔಷಧ: Azithromycin 500mg...",
  "sent_at": "2026-10-08T09:28:05Z"
}
```

Response 200

```
{
  "dispatch_id": "disp_6t3w",
  "status": "sent",
  "message_preview": "ನಿಮ್ಮ ಇಂದಿನ ಸಂಪರ್ಕದ ಸಾರಾಂಶ: ರೋಗನಿರ್ಣಯ – ಗಂಟಲು ಸೋಂಕು. ಔಷಧ: Azithromycin 500mg...",
  "sent_at": "2026-10-08T09:28:05Z"
}
```

8. Doctor Onboarding (Voice Enrollment)

POST /doctors/{doctor_id}/voice-enroll

Enroll a 30-second voice sample so pyannote can identify the doctor's voice in all future sessions.

Content-Type: multipart/form-data

Form fields

Field	Type	Notes
audio	file	.wav , 20-60 seconds, clean speech

Response 201

```
{
  "doctor_id": "d_001",
  "enrollment_status": "complete",
  "voice_profile_id": "vp_d001_v2",
  "quality_score": 0.91,
  "message": "Voice profile created. Speaker identification is now active."
}
```

Error Schema

All errors follow a consistent envelope:

```
{
  "error": {
    "code": "ALLERGY_CONFLICT",
    "message": "Human-readable explanation.",
    "detail": { },
    "request_id": "req_abc123"
  }
}
```

HTTP Status	Meaning
400	Bad request — malformed body
401	Unauthenticated
403	Forbidden — role mismatch
404	Resource not found
409	Conflict — e.g. note already approved
422	Unprocessable — passed validation, failed business rule
500	Internal server error

Webhook Events

For real-time UI updates, the backend emits these events over WebSocket at `wss://api.medassist.local/v1/sessions/{session_id}/stream`:

Event	Payload
<code>transcript.chunk</code>	New diarized lines from latest audio chunk
<code>warning.fired</code>	A rules engine warning was triggered
<code>extraction.complete</code>	Gemini Flash extraction result
<code>note.ready</code>	Draft note is ready for doctor review
<code>note.approved</code>	Final approval confirmed